A counterexample to an additive intersection-index eigenvalue bound

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Abstract

We construct embedded hypersurfaces that contradict the additive intersection-index spectral estimate in AIM 1022. The argument combines a base with large normalized first eigenvalue, tapered connecting handles, a uniform intersection count and Neumann spectral counting.

Repository target. This manuscript concerns AIM problem 1022 at revision `3fa50884d129`. Its scope is stated below.

Theorem. For every integer $n \geq 2$, there do not exist a finite constant $A_n > 0$ and a continuous function $F_n : [0, \infty) \to \mathbb{R}$ such that every smooth compact connected embedded hypersurface Σ^n in $\mathbb{R}^{n+1}$, with finite intersection index, satisfies

$$\lambda_k(\Sigma)\,\mathrm{Vol}_n(\Sigma)^{2/n} \leq F_n(i(\Sigma)) + A_n k^{2/n} \qquad (k \geq 1).$$

The additive question is posed by Pétiard [5, p. 3]. The proof replicates a fixed hypersurface with a large normalized first eigenvalue, joining its copies by long handles whose Neumann spectra contain at most two low modes each. The copies and handles can be placed so that the intersection index stays bounded as their number increases. The fixed base may have a large index; it is independence from the replication number that matters.

The only substantial external inputs are the existence of embedded bases with unbounded normalized first eigenvalue [1, Theorem 1.4], and uniform finiteness for definable families in the o-minimal structure $\mathbb{R}_{\mathrm{an,exp}}$ [2,3]. All handle, placement, smoothing, and spectral-counting details are supplied below. This is a mathematical proof developed in this research session, not a claim of external peer review or priority.

0. Bases with large normalized first eigenvalue and finite definable complexity

Colbois–Dryden–El Soufi [1, Theorem 1.4] prove that, for every $n \geq 2$,

$$\sup_B \lambda_1(B)\,\mathrm{Vol}_n(B)^{2/n} = \infty, \tag{0.1}$$

where B ranges over smooth compact connected embedded hypersurfaces of $\mathbb{R}^{n+1}$. In dimension two the topology may vary; in higher dimensions the result is stronger, but that strengthening is unnecessary here.

We can require each chosen base to be a regular compact component of a polynomial zero set. Here are details of the approximation. A smooth closed embedded hypersurface has a compact tubular neighborhood on which its signed distance rho is smooth, with nonzero normal derivative. Extend rho smoothly to a surrounding compact box. Approximate this extension in C^2 on the box by polynomials. For a sufficiently close polynomial p, its zero set in the tubular neighborhood is a unique C^1-small normal graph over B, with nonvanishing gradient. The polynomial has fixed nonzero signs on the two boundary sheets of a sufficiently small tube, so this zero component cannot leave the tube or join any other zero component. It is therefore a smooth compact connected component of the real algebraic set $p = 0$.

Such a component is semialgebraic. Other components of the polynomial zero set are irrelevant: the base is only the selected compact component. The induced metrics converge uniformly under the C^1 normal-graph convergence. The variational characterization of eigenvalues and the volume formula then give convergence of λ_1 and volume. Hence (0.1) still holds for these semialgebraic bases.

After a Euclidean dilation, normalize

$$\lambda_1(B) = 1.$$

The base volume $V = \mathrm{Vol}_n(B)$ can consequently be arbitrarily large, because this rescaling changes the volume to the $n/2$ power of the original normalized first eigenvalue. Fix one such base for the rest of the construction. Its definable complexity, and all constants depending on it, are allowed to be large but remain fixed as the number N of copies tends to infinity.

A. A straight handle with at most two low Neumann eigenvalues

Fix $a > 0$, length $L > 0$, and radius parameter $\varepsilon > 0$. Put

$$r(s) = \epsilon \frac{\cosh(a(s - L/2))}{\cosh(aL/2)}, \qquad 0 \le s \le L.$$

The n-dimensional tube of revolution in $\mathbb{R}^{n+1}$ has metric

$$g = q(s)^2 ds^2 + r(s)^2 g_{S^{n-1}}, \qquad q(s) = \sqrt{1 + r'(s)^2} \le \sqrt{1 + a^2\epsilon^2} =: Q.$$

With Neumann conditions at the two end cross-sections, separate spherical harmonics. In angular degree zero the Rayleigh quotient is

$$\frac{\int_0^L r^{n-1} q^{-1} |f'|^2 \, ds}{\int_0^L q r^{n-1} |f|^2 \, ds} \ge Q^{-2} \frac{\int_0^L r^{n-1} |f'|^2 \, ds}{\int_0^L r^{n-1} |f|^2 \, ds}. \tag{A1}$$

Write $p = (n-1)/2$. On the codimension-two subspace $H_0^1(0, L)$, putting $\psi = r^p f$ and integrating by parts gives

$$\int_0^L r^{2p} |f'|^2 = \int_0^L \left(|\psi'|^2 + \frac{(r^p)''}{r^p} |\psi|^2 \right),$$

where

$$\frac{(r^p)''}{r^p} = a^2 \left[p \operatorname{sech}^2(a(s - L/2)) + p^2 \tanh^2(a(s - L/2)) \right] \ge a^2 \min(p, p^2). \tag{A2}$$

The min–max principle therefore says that the third angular-zero Neumann eigenvalue (that is, the eigenvalue with index 2 when counting from zero) is at least

$$\Gamma_\epsilon = \frac{a^2 \min(p, p^2)}{1 + a^2\epsilon^2}. \tag{A3}$$

Equivalently, that sector has at most two eigenvalues strictly below Γ_ε. Every nonconstant spherical mode has Rayleigh quotient at least

$$\frac{n-1}{\epsilon^2}, \tag{A4}$$

because its angular eigenvalue is at least $n-1$ and $r \le \varepsilon$. Thus, after fixing a sufficiently large a and then taking epsilon small, **the entire handle has at most two eigenvalues below any prescribed fixed threshold**, independently of its length L.

Its volume also satisfies a length-independent estimate. Since

$$\frac{\cosh(a(s - L/2))}{\cosh(aL/2)} \le e^{-as} + e^{-a(L-s)},$$

and $(x+y)^{n-1} \le 2^{n-2}(x^{n-1} + y^{n-1})$,

$$\mathrm{Vol}_n(\text{handle}) \le Q|S^{n-1}|\frac{2^{n-1}}{(n-1)a}\epsilon^{n-1}. \tag{A5}$$

B. Each straight handle meets an affine line at most four times

If a line has nonconstant axial coordinate s, its transverse coordinate is $A + Bs$. Its intersections with the tube satisfy

$$|A + Bs|^2 - r(s)^2 = 0.$$

The fourth derivative of the left side is strictly negative, since $(r^2)'''' > 0$. Five distinct zeros would contradict Rolle's theorem. If the axial coordinate is constant, the line meets one spherical cross-section in at most two points. Hence the handle contributes at most four transverse intersections to any affine line.

C. A robust bound of eight handles intersected by one line

Let $m = n + 1 \ge 3$. Choose a generic polygonal chain with N vertices in $\mathbb{R}^m$, with all increments in a prescribed open cone around $+e_m$. The vertices may additionally be restricted to an open set with consecutive edge lengths between 1 and 2.

For five **pairwise vertex-disjoint** edges, the set of vertex configurations admitting a common affine-line transversal has positive codimension. Indeed affine lines have dimension $2m-2$; for a fixed line, one segment's endpoints allowing an intersection have dimension at most $m+2$ rather than $2m$. Because the five endpoint pairs are disjoint, these conditions are independent. The total incidence-set dimension is at most

$$mN + (2m-2) - 5(m-2) < mN$$

for every $m \geq 3$. This parametrization also includes endpoint intersections; those strata have no larger dimension. There are only finitely many choices of five edges, so generic vertices avoid all such configurations.

Every collection of nine edges of a chain contains a matching of five pairwise vertex-disjoint edges. Therefore no affine line meets nine edges of this generic chain. A line meets **at most eight** core edges. This slightly weaker constant avoids degeneracies involving adjacent edges and is fully sufficient.

We also choose the vertices so no three are collinear and nonadjacent core edges do not intersect. These are additional finite genericity conditions.

For the fixed finite chain, all these nonintersection conditions have a positive tolerance. A line meeting a fixed bounded neighborhood of the chain belongs to a compact line-parameter space after restricting its offset. Thus, if arbitrarily small tubular neighborhoods of nine chosen edges had a common transversal, compactness would give a transversal of their cores. Taking the minimum over the finitely many subsets gives a positive tube-radius tolerance. The same compactness argument supplies a positive radius tolerance for no line meeting three vertex balls.

Global dilation of the prototype chain increases these margins arbitrarily. Consequently its vertices can be replaced by fixed-size base hypersurfaces, and its edges by arbitrarily long thin handles, while retaining the bounds of two base balls and eight handles per line.

D. Matching the actual ports to the prototype chain

Let the fixed base hypersurface be a regular compact component of a polynomial zero set. Choose a direction v such that both v and -v are regular values of its Gauss map and the height in direction v has a unique global maximum and a unique global minimum. This is possible: Sard's theorem gives the first condition almost everywhere, while almost-everywhere differentiability of the support function of the convex hull gives uniqueness of the two support points almost everywhere. At these extrema, nondegeneracy implies strict convexity of the outward second fundamental form.

The normal maps on small neighborhoods of these two points therefore have analytic semialgebraic inverses

$$p_+(d), \qquad p_-(-d),$$

for d in a sufficiently small open cone around v. The rays leaving these ports in their outward normals avoid the rest of the fixed compact base, after narrowing the cone if necessary.

Let the prototype chain have increments $L_j d_j$. For a dilation factor R, choose actual translations recursively by

$$t_{j+1} = t_j + p_+(d_j) + RL_j d_j - p_-(-d_j). \tag{D1}$$

The outgoing port on copy j and incoming port on copy j+1 are then exactly joined by a straight segment of length RL_j in direction d_j. The actual translations and segments differ from those of the dilated prototype chain by at most a constant times N, because the port offsets are uniformly bounded. For each fixed N, choose R so large that these bounded errors are smaller than the dilated nonintersection margins. This preserves the no-three-base-balls and no-nine-core-edges conclusions. All copies use the same base metric and orientation.

The smooth collars below occupy a fixed-radius base ball, and the straight tube begins a distance epsilon beyond each port. Thus its length is $RL_j - 2\varepsilon$ rather than RL_j; this harmless change is already covered by the estimates in A.

E. Explicit definable collars, without integrating a cutoff

This construction supplies the local family needed for the intersection-index argument. In coordinates centered at a port, with outward normal the z-axis, write the base locally as an analytic semialgebraic graph $z = g(y)$, where $g(0) = 0$ and $Dg(0) = 0$. Its interior lies below the graph.

Choose a fixed nondecreasing C^∞ definable cutoff chi on $[0, 1]$, equal to zero on $[0, 1/4]$ and one on $[3/4, 1]$. For example it can be built from the flat function $\exp(-1/t)$ for $t > 0$, extended by zero for $t \le 0$, using products and quotients; no integration is needed. Put $\eta(t) = t\chi(t)$. Then eta is zero near zero, equals t near one, and has positive derivative when it is not zero.

A fixed corner profile is

$$\rho_0(t) = \epsilon[1 + \eta(1 - t)], \qquad z_0(t) = \epsilon\eta(t), \qquad 0 \le t \le 1.$$

It is planar near t=0 and cylindrical near t=1, with no zero tangent vector. Let χ_{in} be another such cutoff, zero on $[0, 1/2]$ and one on $[3/4, 1]$. Let r be the cosh tube profile, analytically continued a short distance to negative s. Define

$$\rho(t) = \rho_0(t) + \chi_{\text{in}}(t)[r(z_0(t) - \epsilon) - \epsilon].$$

Choose χ_{base} equal to one near zero and zero on $[1/2, 1]$. The collar is parametrized by

$$(t, \theta) \longmapsto (\rho(t)\theta, z_0(t) + \chi_{\text{base}}(t)g(\rho(t)\theta)), \qquad \theta \in S^{n-1}. \tag{E1}$$

Near t=0 this agrees exactly with the base graph on an annulus. Near t=1 it agrees exactly with the continued tube, with axial coordinate $s = z_0(t) - \varepsilon$; thus all derivatives match at both ends. Remove the base disk of radius 2ε and retain the collar up to t=1, where the straight tube begins.

Because $g(y) = O(|y|^2)$ and the cosh correction on a segment of length $O(\varepsilon)$ is $O(a\varepsilon^2)$, after rescaling by epsilon this is a small C^1 perturbation of the fixed corner profile. It is embedded and has a uniformly controlled scaled metric for sufficiently small epsilon. These bounds are uniform in the tube length: on the short continued segment, $|r^{(j)}(s)| \le a^j\varepsilon e^{a\varepsilon}$. The fixed corner profile is a compact embedded annular cylinder with nonzero tangent; sufficiently small C1 perturbations preserve its embedding. One may also choose chi_base nonincreasing, in which case strict convexity of the port graph makes its axial coordinate monotone through the outer transition. The same construction at the other endpoint uses its outward normal, which reverses the tube coordinate; the cosh profile is symmetric about its midpoint.

All functions and parametrizations above are definable in $\mathbb{R}_{\text{an,exp}}$. The base is fixed, its local graph and inverse Gauss maps are semialgebraic, coordinate rotations can be chosen semialgebraically in the normal cone, and the cutoff and cosh terms use only exponentials and algebraic operations. This is a fixed finite-parameter family: parameters are the two port directions, epsilon, a, and the two incident handle lengths L_- and L_+ (with a fixed after the spectral threshold is chosen). A one-port end piece uses only its incident length.

O-minimal uniform finiteness gives a constant $C(\text{base})$ bounding the number of connected components in intersections of any member of this local family with any affine line. For a line transverse to the final smooth hypersurface the intersections are isolated, hence their number is bounded by the same constant. This includes the two collars and the fixed base piece in one base ball. The constant does not depend on N, epsilon, or the lengths of the handles.

Together with C and B,

$$i(\Sigma_N) \le 2C(\text{base}) + 32. \tag{E2}$$

F. Neumann spectra of the base pieces converge to the original base

For a fixed finite collection of port locations, remove the one or two disks of radius 2ε used by the chain and add the collars E1. The first and last copies use one port each; all interior copies use two, and unused ports remain part of the original base. Cut at the tube ends and impose Neumann conditions there. Denote this compact manifold-with-boundary by B_ε.

The standard extension/Rellich argument gives

$$\lambda_1^N(B_\epsilon) \longrightarrow \lambda_1(B), \qquad \mathrm{Vol}(B_\epsilon) \longrightarrow \mathrm{Vol}(B). \tag{F1}$$

Here is the required compactness mechanism, to make clear that no long thin region is hidden in this claim. The collars have diameter $O(\varepsilon)$ and, after scaling, a uniformly controlled fixed geometry. Uniform extension from a Euclidean annulus to its ball, chosen to preserve constants and then transported through local coordinate charts, extends functions on the punctured base to the whole base with uniformly bounded H^1 norm. For clarity, write Γ_ϵ for the common collar/annulus boundary. The fixed-domain trace and Poincaré inequalities, after scaling, give

$$\int_{\text{collar}} |f|^2 \le C\epsilon \int_{\Gamma_\epsilon} |f|^2 + C\epsilon^2 \int_{\text{collar}} |\nabla f|^2,$$

and

$$\epsilon \int_{\Gamma_\epsilon} |f|^2 \le C \int_{\text{annulus}} |f|^2 + C\epsilon^2 \int_{\text{annulus}} |\nabla f|^2.$$

Combining them gives

$$\int_{\text{collar}} |f|^2 \le C \int_{\text{annulus}} |f|^2 + C\epsilon^2 \int_{\text{annulus}} |\nabla f|^2 + C\epsilon^2 \int_{\text{collar}} |\nabla f|^2. \tag{F2}$$

A family with bounded energy and L^2 norm therefore has vanishing L^2 mass in the shrinking collars: after extension, Rellich compactness on the fixed closed base makes the annular L^2 masses vanish. A bounded-energy orthonormal family converges, after subsequences, to an orthonormal family on the base. For the energy lower limit, first apply weak lower semicontinuity on the fixed complement of port balls of radius delta, where the extension agrees with the original function once epsilon is small. Then let delta decrease to zero. This gives the required inequality with the original base-piece energies; the energy introduced inside the artificially filled holes need not tend to zero. Applying min–max to the limiting orthonormal family gives the lower eigenvalue limit. Restriction of smooth test functions, with the obvious uniformly controlled collar extension, gives the upper limit. This works also when n=2; no assertion that a long cylindrical neck has a large gap is used.

For each finite N, only finitely many ports occur. Thus epsilon may be chosen small enough that (F1) holds at every base piece simultaneously. No uniform rate in N is needed.

G. Spectral counting and the contradiction

Normalize the fixed base B so that $\lambda_1(B) = 1$; choose it with volume V as large as needed, using the unbounded normalized first-eigenvalue theorem for smooth embedded hypersurfaces.

Approximation by an algebraic hypersurface can be made C1-small, so this remains possible with a definable base.

Fix a sufficiently large a in A. For each N, construct the chain and then choose epsilon sufficiently small so that every base piece has first positive Neumann eigenvalue at least 1/2, volume at least $V/2$, and every handle has at most two Neumann eigenvalues below 1/2.

The construction replaces exactly two disk boundaries for every connecting tube, so the result is a smooth closed hypersurface. The chain makes it connected, and the placement and clearance estimates make it embedded. Cut this hypersurface along all base/handle interfaces. Its Neumann-bracketed disjoint union has at most

$$N + 2(N-1) = 3N - 2$$

eigenvalues strictly below 1/2, counting its zero modes. Since eigenvalues are indexed from zero, min–max gives

$$\lambda_{3N-2}(\Sigma_N) \geq \frac{1}{2}, \qquad \mathrm{Vol}_n(\Sigma_N) \geq \frac{NV}{2}. \tag{G1}$$

The intersection index is bounded independently of N by E2. If the proposed additive estimate held, continuity of F on the compact interval $[0, 2C(base) + 32]$ would bound its first term by a finite number M_B. Thus

$$\frac{1}{2}\left(\frac{NV}{2}\right)^{2/n} \leq M_B + A_n(3N-2)^{2/n}.$$

Divide by $N^{2/n}$ and let N tend to infinity:

$$\frac{1}{2}\left(\frac{V}{2}\right)^{2/n} \leq A_n 3^{2/n}.$$

Since the normalized base volume V can be arbitrarily large, this is impossible for a dimension-only finite A_n.

The exact numerical constants 2, 8, 32, and 3 are irrelevant; their independence of N is what permits the contradiction.

H. Two strengthenings

Continuity of the proposed additive term is not needed for this contradiction. The intersection index is an integer and stays in the finite set $\{0, 1, \ldots, \lfloor 2C(\mathrm{base}) + 32 \rfloor\}$. Any real-valued function on the nonnegative integers is bounded on that finite set. Thus the same argument rules out an arbitrary finite-valued additive function of the index.

For $n \geq 3$, the counterexample can also be confined to hypersurfaces diffeomorphic to S^n. In Theorem 1.4(2) of [1], take the fixed original submanifold to be the standard embedded sphere. The selected base is then an n-sphere, and the algebraic approximation preserves its diffeomorphism type. Joining N such components along a chain is their connected sum, again diffeomorphic to S^n. The large normalized first eigenvalue input in dimension two requires varying topology, so this additional fixed-topology conclusion is asserted only for $n \geq 3$.

References

[1] B. Colbois, E. B. Dryden, A. El Soufi, *Bounding the eigenvalues of the Laplace–Beltrami operator on compact submanifolds*, Bulletin of the London Mathematical Society **42** (2010), 96–108, Theorem 1.4. DOI: https://doi.org/10.1112/blms/bdp100. Primary preprint: https://arxiv.org/abs/0909.5346.

[2] L. van den Dries, A. Macintyre, D. Marker, *The elementary theory of restricted analytic fields with exponentiation*, Annals of Mathematics **140** (1994), 183–205. DOI: https://doi.org/10.2307/2118545. This establishes o-minimality of the restricted analytic structure expanded by the real exponential function.

[3] L. van den Dries, C. Miller, *On the real exponential field with restricted analytic functions*, Israel Journal of Mathematics **85** (1994), 19–56. The uniform finite-component bound used in Section E is the standard consequence of definable cell decomposition: for a fixed definable family, the number of connected components of every fiber is bounded uniformly over its parameters. A precise statement is Section 4.4 (PDF p. 13) of [4].

[4] L. van den Dries and C. Miller, *Geometric categories and o-minimal structures*, Duke Mathematical Journal **84** (1996), 497–540; author's revised manuscript dated 20 February 2001. Section 4.4 gives uniform bounds on connected components of fibers. https://people.math.osu.edu/miller.1987/newg.pdf. Only this uniform-fiber result is used here.

[5] L. Pétiard, *Counter-example to a Kröger type spectral inequality*, arXiv:1706.09189v1 (2017), six-page preprint. The additive intersection-index question is on p. 3, immediately before Section 2. https://arxiv.org/abs/1706.09189v1.

The polynomial approximation, min–max principle, coarea-free one-dimensional handle estimates, and elementary extension/Rellich argument are presented in the proof. No numerical approximation is used.